

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE (SPECIAL) EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 10—COMPLEX ANALYSIS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 10—COMPLEX ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Real part of $f(z) = \log z$ is :

- (A) $\frac{1}{2} \log(x^2 + y^2).$ (B) $\log(x^2 + y^2).$
(C) $\log(x + iy).$ (D) None of these.

2. If $f(z) = (x^2 + ay^2 - 2xy) + i(bx^2 - y^2 + 2xy)$ is analytic, then value of a and b is :

- (A) $-1, 1.$ (B) $1, -1.$
(C) $1, 0.$ (D) None of these.

3. If S and T are domains in the complex plane, which of the following need NOT be true ?

- (A) $S \cup T$ is a domain if $S \cap T \neq \phi.$ (B) $S \cup T$ is an open set.
(C) $S \cap T$ is an open set. (D) $S \cap T$ is a domain.

4. If $f(z)$ is a real valued analytic function in a domain D , then :

- (A) $f(z)$ is a constant. (B) $f(z)$ is identically zero.
(C) $f(z)$ has modulus 1. (D) None of these.

5. For real numbers x and y , $\sin(x + iy)$ equals :

- (A) $\sin x \cosh y + i \cos x \sinh y.$ (B) $\cos x \cosh y - i \sin x \sinh y.$
(C) $\sin x \cosh y - i \cos x \sinh y.$ (D) $\cos x \cosh y + i \sin x \sinh y.$

6. An arc $z = z(t); a \leq t \leq b$ is simple if :

- (A) $z(t)$ is continuous. (B) $z(t)$ is a one to one function.
(C) $z(t)$ is such that $z(a) = z(b).$ (D) None of these.

7. The only bounded entire functions are :

- (A) Real valued functions. (B) Harmonic functions.
(C) Constant functions. (D) Exponential function.

8. Suppose $f(z)$ is analytic inside and on unit circle. If $|f(z)| \leq 1, \forall z$ with $|z| = 1$. Then an upper bound for $|f''(0)|$ is :
- (A) $n!$. (B) $\frac{1}{n!}$.
 (C) n . (D) None of these.
9. Value of the integral $\int_0^\pi e^{it} dt$ is :
- (A) $2i$. (B) 0 .
 (C) $2\pi i$. (D) None of these.
10. If n is any non-zero integer, then $\int_0^{2\pi} e^{in\theta} d\theta$ equals :
- (A) 0 . (B) 2π .
 (C) 1 . (D) None of these.
11. A Maclaurin series is a Taylor series with centre :
- (A) $z_0 = 1$. (B) $z_0 = 0$.
 (C) $z_0 = 2$. (D) None of these.
12. The radius of convergence of the power series of the function $f(z) = \frac{1}{1-z}$ about $z = 1/4$ is :
- (A) 1 . (B) $1/4$.
 (C) $3/4$. (D) 0 .
13. A power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ always converges for :
- (A) At least one point z .
 (B) All complex numbers z .
 (C) At all z which are either real or purely imaginary.
 (D) At all z with $|z - z_0| < R$ for some $R > 0$.
14. The singular points of the function $f(z) = \frac{1}{4z - z^2}$ are :
- (A) $z = 0$ and $z = -4$. (B) $z = 0$ and $z = 4$.
 (C) $z = 4$ and $z = -4$. (D) $z = 2$ and $z = -2$.

Turn over

15. The constant term in the Laurent series expansion of $f(z) = \frac{e^z}{z^2}$ in the region $0 < |z| < \infty$ is:

- (A) 0. (B) $\frac{1}{2}$.
(C) 2. (D) None of these.

16. Value of $\int_0^{\infty} \frac{\sin x}{x} dx$ is:

- (A) $\frac{\pi}{2}$. (B) $\frac{1}{2}$.
(C) ∞ . (D) 2π .

17. If $f(z)$ has a pole of order m at z_0 , then at z_0 , $\frac{1}{f(z)}$ has:

- (A) A removable singularity. (B) An essential singularity.
(C) A pole of order m . (D) None of these.

18. For $f(z) = \frac{\tan z}{z}$, $z = 0$ is a:

- (A) Essential singularity. (B) Simple pole.
(C) Removable singularity. (D) Double pole.

19. Singularities of a rational function are:

- (A) Poles. (B) Essential.
(C) Non-isolated. (D) Removable.

20. The singularity of the function $\frac{\sin z}{z}$ at $z = 0$ is:

- (A) Essential singularity. (B) Simple Pole.
(C) Removable singularity. (D) Double Pole.

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE (SPECIAL) EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 10—COMPLEX ANALYSIS

Time : Three Hours

Maximum : 120 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. Define Harmonic functions.
2. Is the function $f(z) = \bar{z}$ analytic on $\mathbb{C}$.
3. Define simply connected domain.
4. Write Cauchy-Riemann equations in polar co-ordinates.
5. If a function f fails to be analytic at a point z_0 but is analytic at some point in every neighborhood of z_0 , then z_0 is called a _____ of f .
6. State Cauchy's integral formula.
7. What is the principal value of $(-i)^i$?
8. Write an upper bound of $\sin z$, $z \in \mathbb{C}$.
9. Define *simple connected* domain.
10. State Maximum modulus principle.
11. Give an example of non-isolated singular point.
12. If $\lim_{z \rightarrow z_0} f(z) = \infty$, then z_0 is called _____ of f .

(12 × 1 = 12 marks)

Turn over

Section B

Answer at least **eight** questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. Define entire functions. Give an example.
14. Briefly explain branch of logarithm. Also mention about *principal branch*, *branch cut* and *branch point*.
15. Show that $\text{Log}(1 - i) = \frac{1}{2} \ln(2) - \frac{\pi}{4}i$.
16. Find the principal value of i^i .
17. Evaluate $\int_C (z + 2)/z dz$, where C is the positively oriented semicircle $|z| = 2$ in the upper half plane.
18. State Cauchy's integral formula and its extension.
19. Show that the derivative of an analytic function is analytic.
20. Suppose that $z_n = x_n + iy_n$ ($n = 1, 2, \dots$) and $S = X + iY$. Then show that $\sum_{n=1}^{\infty} z_n = S$ if and only if $\sum_{n=1}^{\infty} x_n = X$ and $\sum_{n=1}^{\infty} y_n = Y$.
21. Prove or disprove : If f is an analytic function on a connected domain D and C is a simple closed curve in D, then $\int_C f(z) dz = 0$.
22. Suppose $z_n = x_n + iy_n$, $n = 1, 2, 3, \dots$, $x_n \rightarrow x$ and $y_n \rightarrow y$ then show that $z_n \rightarrow z$, where $z = x + iy$.
23. Assuming the Maclaurin series of $\sin z$, obtain the Maclaurin series of $\sinh z$.
24. Discuss about the residue of an analytic function at infinity.
25. Find the residue of $\frac{1}{z + z^2}$ at $z = 0$.
26. Show that the zeros of a non-zero analytic function are isolated.

(8 × 6 = 48 marks)

Section C

Answer at least **five** questions.
Each question carries 9 marks.
All questions can be attended.
Overall Ceiling 45.

27. Derive Cauchy-Riemann equations.
28. Show that $\sin^{-1}(z) = -i \log \left[iz - (i - z^2)^{1/2} \right]$.
29. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be analytic. Define $g: \mathbb{C} \rightarrow \mathbb{C}$ by $g(z) = \overline{f(\bar{z})}$. Is g analytic? Prove your claim.
30. Prove Liouville's theorem, and hence deduce the fundamental theorem of algebra.
31. State and prove the principle of domination of paths.
32. Suppose that $|f(z)| \leq |f(z_0)|$ at each point z in some neighborhood $|z - z_0| < \epsilon$ in which f is analytic. Then show that $f(z)$ has the constant value $f(z_0)$ throughout that neighborhood.
33. $f(z_1)$ is a point inside the circle of convergence $|z - z_0| = R$ of a power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$, then show that the series must be uniformly convergent in the closed disk $|z - z_0| \leq R_1$, where $R_1 = |z_1 - z_0|$.
34. Explain the role of residues in the evaluation of real improper integral.
35. Show that $\int_0^{2\pi} \frac{d\theta}{1 + a \sin \theta} = \frac{2\pi}{\sqrt{1 - a^2}}$.

(5 × 9 = 45 marks)

Section D

Answer any **one** question.
The question carries 15 marks.

36. (a) Define conjugate harmonic function. Show that $u(x, y) = 2x(1 - y)$ is harmonic and find its harmonic conjugate.
- (b) If u is harmonic and v is harmonic conjugate to u , then show that $u + iv$ is analytic.

Turn over

37. Define absolute convergence of a power series. Also show that if a power series $\sum_{n=1}^{\infty} a_n (z - z_0)^n$ converges when $z = z_1$ ($z_1 \neq z_0$) then it is absolutely convergent at each point z in the open disk $|z - z_0| < |z_1 - z_0|$.

38. Evaluate $\int_0^{\infty} \frac{x^2}{x^6 + 1} dx$.

(1 × 15 = 15 marks)

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